

Electro Mechanical Systems (EE2010)

Date: April 09th, 2025

Course Instructor(s)

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Sessional-2 Exam

Total Time: 1 Hours

Total Marks: 50

Total Questions: 02

Semester: SP-2025

Campus: Lahore

Dept: Electrical
Engineering

Student Name

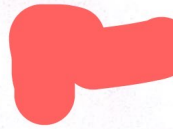
Roll No

Section

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**CLO #1:** Investigate working of a DC Machine**Q1:****[marks =25]**

A 30HP, 240V compensated DC shunt motor has armature resistance of 0.35Ω . Its field resistance is 90Ω and adjustable resistance connected in series with field coil can vary from 70Ω to 300Ω . Its no load characteristic curve at a speed of 1400 rpm is tabulated as under.

I_F (A)	0.3	0.6	0.9	1.2	1.4	1.6
E_A (V)	130	230	274	288	292	295

Investigate the DC machine to find out following quantities assuming rated terminal voltage in all cases.

- What is the maximum and minimum no load speed at which this motor can operate
- No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 120 ohm.
- The induced torque in loaded condition at line current of 70A, with same field current as in part (b)
- Fully labeled Circuit diagram for part (c)
- Also draw the general torque speed curve of a compensated DC shunt motor

CLO #3: Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation**Q2:****[marks 25]**

A 2300 V, 1200 KVA, 0.85-PF-lagging, 50-Hz, 24 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 28.5 KW, and its core losses are 23.5 KW. The field circuit has a dc voltage of 220 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 12 to 200 Ω . The generator is providing 250 A to a load at rated power factor and a terminal voltage of 2200V. **Analyze** the Synchronous Generator circuit to find out the following quantities

- Rotational speed
- Internally generated voltage
- Voltage regulation
- Efficiency
- Output torque of prime mover